Happy Hoppy Frogs
Development Office
Spring 2025

Element E: Application of STEM Principles and Practices

The development office at our independent boarding and day school has an overflow of items. The excess items piled on the floor hinder workflow and efficiency. Many items move around the development office; therefore, it is important to keep them organized. The current storage situation is unsatisfactory in the eyes of school donors and visitors.

We can use measurements to make sure the outlets stay accessible after we install the solution and that the boxes fit well on the shelves. Accurate measurements will also help ensure the boxes fit in smoothly and work properly in the solution.

To conceal the boxes, we will include doors on two sections of our solution. Hinges will allow the doors to open and close smoothly, providing seamless access while keeping the boxes hidden when not in use. This design encompasses simple yet effective technology to ensure functionality and maintain an organized, clean appearance (Types, uses, components, and considerations of hinges).

Shear, moment, and load distribution help predict whether our solution can support the heavy boxes. One team member brings some expertise in these physics concepts, having completed advanced physics courses and gained further insights from a professional engineer. Our research confirms that shear, moment, and load distribution play a critical role in any design (Relation between loading, shear and moment, 2021). These principles allow us to ensure the structure remains stable and can reliably support the intended load.

Furthermore, our design was reviewed by three experts with experience in woodworking, engineering, and construction. Director of the Performing Arts at our independent private boarding and day school, Mr. C., who has substantial construction experience, and frequently helps with set design for drama productions, confirmed our use of structural principles and offered input on the practicality of our material choices. Mr. H, a woodworking teacher, validated the measurements and proportions, emphasizing the need for additional support to prevent bowing. Mr. M, a professional engineer, reviewed our design from a safety and engineering perspective, supporting our application of STEM concepts and suggesting structural improvements to reinforce stability.

Mr. C., Mr. H., and Mr. M. all strongly recommended using ¾" sanded plywood because it is sturdy, reliable, and can handle stress well. Since all three experts agreed, we will definitely use this material as the base for the project. Mr. C. suggested adding trim to cover the raw edges of the plywood, making the workspace look smoother and more polished. Mr. H. emphasized the need for extra bracing to make the structure stronger. He also advised painting the shelf, which will make it look better and help protect it from damage. Mr. M. pointed out that the upper shelving needs backing support to prevent bending over time.

We are making several changes based on these recommendations. Adding bracing will make sure the structure stays stable. Using ¾" plywood will give the workspace the strength it needs, while keeping our budget in mind. Applying trim will improve its appearance, and adding backing to the upper shelves will stop them from bending.

References

Types, uses, components, and considerations of hinges. (n.d.). Retrieved April 25, 2025, from https://www.iqsdirectory.com/articles/hinge.html

8. 6: Relation between loading, shear and moment. (2021, August 14). Engineering LibreTexts. https://eng.libretexts.org/Bookshelves/Mechanical_Engineering/Engineering_Statics%3A

_Open_and_Interactive_(Baker_and_Haynes)/08%3A_Internal_Loadings/8.06%3A_Relation_Between_Loading_Shear_and_Moment

https://www.calcbook.com/post/understanding-shear-and-moment-diagrams-in-structural-engineering